G

g, *see* Ground state
G values, for water radiolysis, 220–221, 221t, 222f
Gamma camera, 578f, 579f, 580, 581t, 582t, 584f, 585f
Gamma decay, 86–87, 97–99, 98f, 99f
  classification of, 153t
  decay energy of, 98
  detection of, 98
  in electron neutrino detection, 23–25
  isomers and, 32
  nuclear structure and, 152–154, 153t
  properties of, 21t
  terminology for, 101–102
Gamma radiography, 203–204, 204t
Gamma radiolysis, 228–229
Gamma spectrometry, 279–283, 280f, 282f
Gamma(γ)-radiation, 4
  absorption of, 182–188
    attenuation coefficient, 182–183, 183f
    coherent scattering, 184, 184f
    Compton effect, 184f, 187–188, 188f
    dominance regions for, 188, 188f
    partial absorption process, 183
    photoelectric effect, 184–187, 184f, 185f
  americium for, 440–441
  in biological systems, 450, 450t
  californium for, 441
  cancer frequency and, 474, 474f
  with fission, 331
  inorganic substance exposure to, 218
  organic compounds and, 227, 227t
  in photodisintegration, 337
  positron emission and, 178–179
  tracks with, 213
  from uranium ores, 388
  in water, 213f, 233
Gamow Teller selection rules for beta decay, 154, 154t
GANIL accelerator complex, 504–505, 506f
Gas centrifugation, 59–60, 59f
  stationary-walled, 56f, 60–61
Gas counters, 254–262, 255f, 256f
  Geiger–Müller counters, 90, 254, 255f, 256f, 260–262, 260t
  ion chambers, 255–258, 256f
  proportional counters, 88–90, 254, 256f, 258–260, 259f, 260t
Gas multiplication factor
  in ion chambers, 254, 255f
  in proportional counters, 258
Gas scintillator detectors, 271
Gaseous diffusion, 57–58
Gas-jet transportation systems, 537–538
Gas-phase separation, of radionuclides, 537–539, 538f–539f
GCR, *see* Magnox reactor
Geiger–Müller (GM) counters, 90, 254, 255f, 256f, 260–262, 260t
General labelling, radioactive, 528
Genetic effects of radiation, 467
Geologic formations
  HLWs deposition in, 741–747
    repository projects, 740f, 743f, 747
    waste conditioning before, 740f, 743f, 745–746
  properties of, 740f, 742–745
Geometric efficiency (ψ_{geom}), 166–167, 166f
  for absolute disintegration rates, 285
Geometrical buckling, 611–612
Giant resonance region, 337
Girdler-Sulphide (G-S) process, 55, 56f
Glass, radiation and, 218–219
Glass dosimeter, 231
Gluon, 17t, 18, 26
GM, *see* Geiger–Müller
GM counters, *see* Geiger-Müller
Graphite moderated and water cooled
  history of, 12
  plutonium formation in, 699–700, 701t
Gravitational microlenses, 343–344
Graviton, strength and range of, 17–18, 17t
Gravity
  escape velocity and, 343
  photons and, 19
  solar system and, 26
  strength and exchange particles of, 17–18, 17t
  weak interaction force compared with, 18–19
Ground state (g), 32–33
G-S process, *see* Girdler-Sulphide process
Gyromagnetic ratio, 134, 137

H

Ha, *see* Hassium
Hadrons, quarks of, 27

Half-life, 4, 86
of actinide elements, 439t
of alpha decay, 155, 157, 158f
in body, 479
decay energy and, 119
metastable and, 32–33
notation for, 6–7
of promethium, 79
for spontaneous fission, 315, 334f
Halide volatility, 709
Hassium (Ha), 440t, 442
Head end plant, 710–713, 711f
Health chemists, 489
Health physicists, 446, 489
Heat conductivity, of fuel, 692, 693f
Heat of formation (ΔH), 70–71
molecular stability and, 72
Heavy water
neutron moderation with, 606t, 607
production of
with chemical exchange, 55, 56f
cryogenic distillation, 60
with electrolysis, 56
Heisenberg uncertainty principle, 120–121
Helium (He)
binding energy for, 72
burning to iron, 358–361, 359f, 360f
cloud chamber pictures of, 15, 16f
hydrogen fusion into, 347, 354–358, 356f, 356t, 357t
isotopic data for, 34t
mass defect of, 72
Helium cooled and graphite-moderated reactor (HTTR), 663–664
Helium flash, 345f, 358
Hertzsprung–Russell (HR) diagrams, 344–345, 345f
Heterogeneous reactors, 600
HF, *see* Hindrance factor
hfs, *see* Hyperfine spectra
High-level liquid wastes (HLLWs)
solidification of, 737f, 739–741, 739f, 741t
tank storage of, 729f, 730–731
High-level wastes (HLWs)
beneficial utilization of, 748–749
deposition in geologic formations, 741–747
repository projects, 740f, 743f, 747
waste conditioning before, 740f, 743f, 745–746
final treatment of, 731–738
dispersion into sea and air, 730f, 731–734, 733f
disposal into space, 737–738
partitioning of, 733f, 734f, 735–736, 736f
remote deposition, 738
transmutation, 738
High-purity germanium (HPGe) detectors, 268, 268f
efficiency curves for, 281, 282f
typical peak from, 282–283, 282f
Hindrance factor (HF), 157, 159–161
HLLWs, *see* High-level liquid wastes
HLWs, *see* High-level wastes
Hold-back carrier, 550, 570
Hole theory, 23
Homogeneous reactors, 600
Hot atom chemistry, 531
Hot atom reactions, 531–532
Hot fusion, 429
HPGe detectors, *see* High-purity germanium detectors
HR diagrams, *see* Hertzsprung-Russell diagrams
HTTR, *see* Helium cooled and graphite-moderated reactor
Hubble constant, 345–346
Hubble law, 345–346
Hydrogen (H)
fusion to helium, 347, 354–358, 356f, 356t, 357t
isotope effect in, 41–42
isotopic data for, 34t
isotopic line shifts for, 42–43
in nuclear reactor water, 222
optical emission spectra for, 42–43
in water cooled reactors, 668
Hydrogen bombs, 648–649, 650f
tritium from, 379
Hydrolysis, of actinide elements, 766f, 767–769
Hydroxyl radical, 223
Hyperfine spectra (hfs), 148–149, 149f

I

Immunoassay, radioactive tracers in, 575
INAA, *see* Instrumental NAA
Industrial radiation processing, 234–235
Inelastic scattering, 297, 306–307